FOUR EQUIVALENT GUISES OF FINITENESS FOR MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7847-7860 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7862-7900, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete); spaces proof materially uses the chapter's etale-splitting machinery.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms of Spaces: finite-proper, finite-quasi-finite, locally-quasi-finite, universally-closed-quasi-compact, quasi-finite-at-point, integral-local, image and point lemmas of Properties of Spaces; chapter lemma etale-splits-off-quasi-finite-part.

Source reference. Original lemma 02LP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7472–7482. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03ZQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06RW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 06LS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03H4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7900–7900. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

1. *f is finite,*
2. *f is proper and locally quasi-finite,*
3. *f is proper and $|X_k|$ is a discrete space for every morphism $\mathrm{Spec}(k) \to Y$ where k is a field,*
4. *f is universally closed, separated, locally of finite type and $|X_k|$ is a discrete space for every morphism $\mathrm{Spec}(k) \to Y$ where k is a field.*

Proof. We have (1) $\Rightarrow$ (2) by Morphisms of Spaces, Lemmas 04NZ, 04NY. We have (2) $\Rightarrow$ (3) by Morphisms of Spaces, Lemma 06RW. By definition (3) implies (4).

Assume (4). Since f is universally closed it is quasi-compact (Morphisms of Spaces, Lemma 04XW). Pick a point y of $|Y|$. We represent y by a morphism $\mathrm{Spec}(k) \to Y$. Note that $|X_k|$ is finite discrete as a quasi-compact discrete space. The map $|X_k| \to |X|$ surjects onto the fibre of $|X| \to |Y|$ over y (Properties of Spaces, Lemma 03H4). By Morphisms of Spaces, Lemma 06LS we see that $X \to Y$ is quasi-finite at all the points of the fibre of $|X| \to |Y|$ over y. Choose an elementary étale neighbourhood $(U, u) \to (Y, y)$ and decomposition $X_U = V \amalg W$ as in Lemma 02LP adapted to all the points of $|X|$ lying over y. Note that $W_u = \emptyset$ because we used all the points in the fibre of $|X| \to |Y|$ over y. Since f is universally closed we see that the image of $|W|$ in $|U|$ is a closed set not containing u. After shrinking U we may assume that $W = \emptyset$. In other words we see that $X_U = V$ is finite over U. Since $y \in |Y|$ was arbitrary this means there exists a family $\{U_i \to Y\}$ of étale morphisms whose images cover Y such that the base changes $X_{U_i} \to U_i$ are finite. We conclude that f is finite by Morphisms of Spaces, Lemma 03ZQ. $\square$